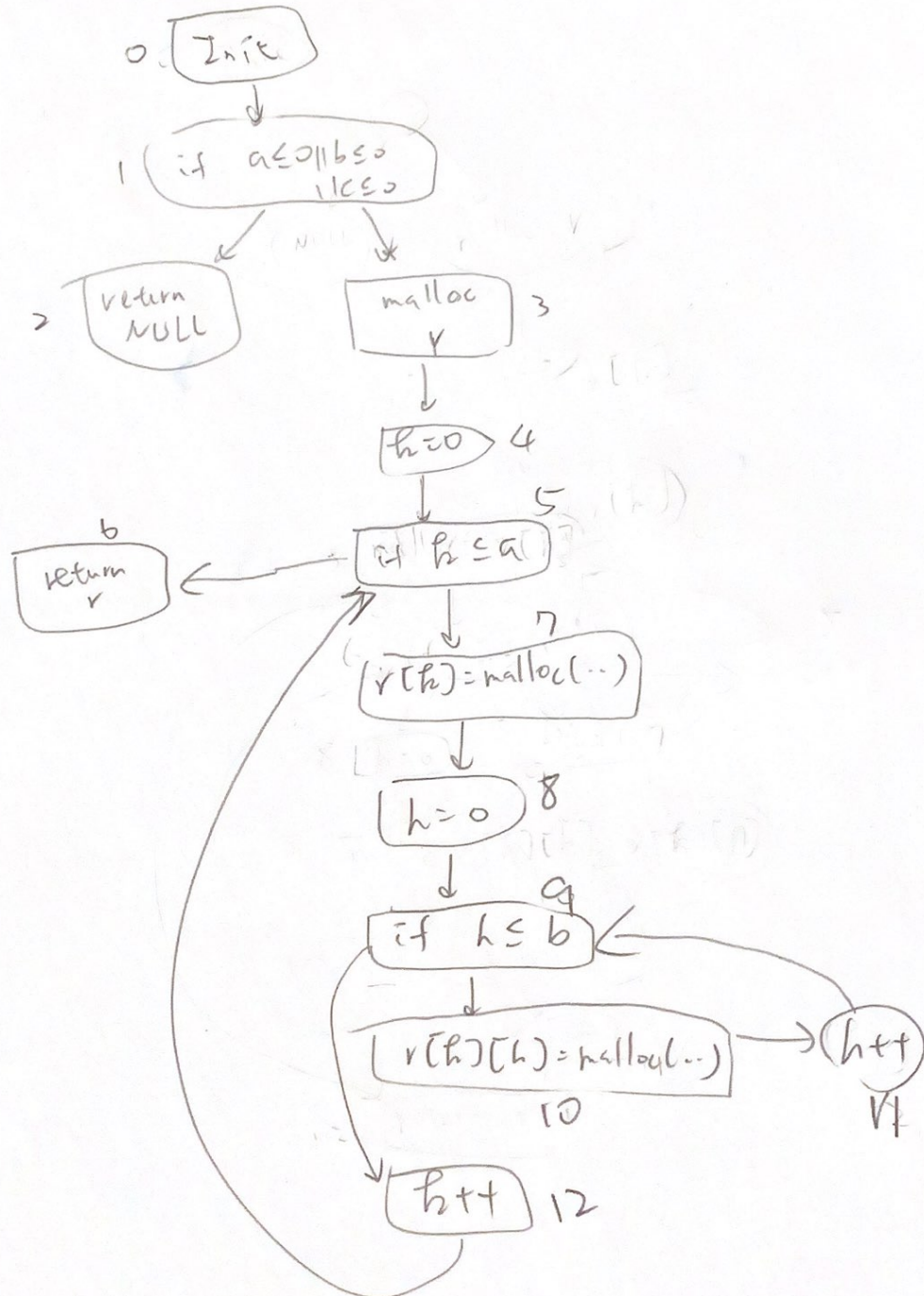


1. Draw the control-flow graph of the above program. (5pts/5)



2. List the test requirements for prime-path coverage (PPC) continuing from question 1. (5pts/10)

List all 'prime path'

① 1 → 2

② 1 → 3 → 4 → 5 → 7 → 8 → 9 → 10 → 2 → 1

~~③ 5 → 6~~

④ 9 → 10 → 11 → 9

⑤ 10 → 11 → 9 → 12 → 5 → 6

⑥ 10 → 11 → 9 → 12 → 5 → 7 → 8 → 9 → 10

3. Continuing from question 2, please write down a test set (i.e, values of a,b,c) for the full PPC. (5pts/15)

(0, 0, 0)

(1, 1, 1)

(2, 1, 1)

(, ,)

4. Continuing from question 2, please list all of the du-paths with respect to a , b , c , and r . (Note: Include all du-paths, even those that are subpaths of some other du-path). (5pts/20)

$$d_r = \{3, 7, 10\}, U_r = \{6, 7, 10\}$$

$$d_a = \{0\}, U_a = \{1, 5, 3\}$$

$$d_b = \{0\}, U_b = \{1, 9, 7\}$$

$$d_c = \{0\}, U_c = \{1, 10\}$$

For r , $3 \rightarrow 4 \rightarrow 5 \rightarrow 7$ $10 \rightarrow 11 \rightarrow 12 \rightarrow 5 \rightarrow 6$
 $3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 10$

For a , $0 \rightarrow 1$
 $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5$
 $0 \rightarrow 1 \rightarrow 3$

For b , $0 \rightarrow 1$
 $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 8 \rightarrow 9$
 $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7$

For c , $0 \rightarrow 1$
 $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 10$

5. Continuing from question 4, please write down a minimal test set (i.e., values of a, b, c) that satisfies all-defs coverage with respect to a , b , c , and r . (Direct tours only). (5pts/25)

$(2, 2, 2)$

$(1, 1, 1)$

6. Continuing from question 4, please write down a minimal test set (i.e, values of a, b, c) that satisfies all-uses coverage with respect to a, b, c , and r . (Direct tours only.) (5pts/30)

(2, 2, 2)

(1, 1, 2)

$$A + B(C + D)$$

7. Consider the predicate $a < 2 \parallel ((!p) \&\& (r \parallel s \geq t))$. Please compute (and simplify) the conditions under which each of the clauses (atoms in standard logic terms) that determines this predicate. (5pts/35)

$$\begin{aligned} (a < 2, r, s \geq t) &= (T, F, F) \\ &\quad (T, F, T) \\ &\quad (T, T, F) \Rightarrow a < 2 \text{ determines} \\ &\quad (T, T, T) \\ &\quad (F, F, F) \end{aligned}$$

$$\begin{aligned} (a < 2, r, s \geq t) &= (F, T, T) \\ &\quad (F, T, F) \Rightarrow p \text{ determines} \\ &\quad (F, F, T) \end{aligned}$$

$$(a < 2, p, s \geq t) = (F, F, F) \Rightarrow v \text{ determines}$$

$$(a < 2, p, r) = (F, F, F) \Rightarrow s \geq t \text{ determines}$$

$$\text{let } A = a < 2, B = p, C = r, D = s \geq t$$

$$\begin{aligned} P_A &\equiv (T + \overline{B}(C + D)) \oplus (F + \overline{B}(C + D)) \\ &\equiv \overline{B}(C + D) \equiv B + \overline{(C + D)} \equiv B + \overline{CD} \end{aligned}$$

$$\begin{aligned} P_B &\equiv (A + (T)(C + D)) \oplus (A + (F)(C + D)) \\ &\equiv (A + C + D) \oplus A \end{aligned}$$

$$\begin{aligned} P_C &\equiv (A + B(T + D)) \oplus (A + B(F + D)) \\ &\equiv (A + B) \oplus (A + BD) \end{aligned}$$

$$\begin{aligned} P_D &\equiv (A + B(C + T)) \oplus (A + B(C + F)) \\ &\equiv (A + B) \oplus (A + BC) \end{aligned}$$

8. Continuing from question 7, write the complete truth table for all clauses (atoms in standard logic terms). (5pts/40)

IC	$\bar{f}$	$a < 2$	p	r	$s \geq t$	predicate	$\neg(a < 2)$	$\neg p$	$\neg r$	$s < t$
	T	F	F	F	F	F	✓		✓	✓
	T	F	F	F	T	T		✓		✓
	F	F	F	T	F	T		✓	✓	
	F	F	F	T	T	T		✓		
	T	F	T	F	F	F	✓			
	T	F	T	F	T	F	✓			
	T	F	T	T	F	F	✓	✓		
	T	F	T	T	T	F	✓	✓		
	F	T	F	F	F	T	✓			
	F	T	F	F	T	T	✓			
	F	T	F	T	F	T	✓			
	F	T	F	T	T	T	✓			
	F	T	T	F	F	T	✓	✓		
	F	T	T	F	T	T	✓	✓		
	F	T	T	T	F	T	✓	✓		
	F	T	T	T	T	T	✓	✓		

9. Continuing from question 8, please identify all pairs of rows from your truth table that satisfy general active clause coverage (GACC) with respect to each clause (atoms in standard logic terms). (5pts/45)

$$\begin{array}{l} \text{GACC} \\ \begin{array}{l} (FFFF)(TFFF) \quad (FFFT)(FTFT) \\ (FTFF)(TTFF) \quad (FFTF)(FTTF) \\ (FTFT)(TTFT) \quad (FFTT)(FTTT) \\ (FTTF)(TTTF) \\ (FTTT)(TTTT) \end{array} \\ \hline (FFFF)(FFTF), (FFFF)(FFFT) \end{array}$$

10. Continuing from question 8, identify all rows from your table that satisfy *implicant coverage (IC)* with respect to each implicant. (5pts/50)

$$\begin{aligned} f &= (a < 2) \vee (\neg p \wedge (r \vee (s < t))) \\ \bar{f} &= \neg(a < 2) \wedge (p \vee (\neg r \wedge \neg(s < t))) \\ &\quad \text{TFTT} \\ f &= A + \bar{B}(C + D) = A + \bar{B}C + \bar{B}D \\ \bar{f} &= \bar{A}(\bar{B}(C + D)) \\ &= \bar{A}(\bar{B} + \overline{(C + D)}) \quad \text{FTTT} \\ &= \bar{A}(\bar{B} + \bar{C}\bar{D}) = \bar{A}\bar{B} + \bar{A}\bar{C}\bar{D} \\ \text{Implicants} &= \{A, \bar{B}C, \bar{B}D, \bar{A}\bar{B}, \bar{A}\bar{C}\bar{D}\} \\ \text{IC} &= \{(TFTT), (FTTF)\} \end{aligned}$$

11. Assume that we have a web program interface for 2-step verification payment with the following screen layout.

Credit card number: 16 digits X

Valid through: mm/yy

OTP: 6 digits cvc: 3 digits

email: @

pay

Send OTP via email!

Figure 2: a GUI for registration and query password.

There is no other document. Please define characteristics of inputs to the web interface and list them. (5pts/55)

12. Continuing from question 11, please present a fuzz testcase generation algorithm for the text fields according to the characteristics you defined. Please argue for your fuzzing scheme. (5pts/60)

fuzzing → 亂猜的
 Credit card num: random 16 characters
 valid through: 01/5/13 x 01/5/19 (13/19 or expired)

OTP: random 6 characters

cvc: random 3 characters

email: rand(10, char) @ rand(10, char) . (15)

↑
 repeat
 "random"
 times

13. In software testing, what is the difference between verification and validation?
(3pts/63)

verification $\rightarrow$ fit requirement
達成規格

validation $\rightarrow$ intended use
使用目的

14. In software quality control, what is static testing? What is dynamic testing?
(2pts/65)

static : test without code execution

dynamic : test under execution

15. What are the five levels of test process maturity? (5pts/70)

lv 1 : testing = debugging

2 : aim to show correctness

3 : aim to show failure

4 : aim to reduce risk

5 : mental discipline that
improves software quality

16. Assume that we have the procedure in Figure 1 to test. Please write down a mutant with the Arithmetic Operator Replacement (AOR) and write down a test case (values for a, b, and c) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/75)

mutant

```
if (a > 0) || b <= 0 || c <= 0
    return NULL;
```

no mutant

(a, b, c) = (0, 1, 1)

mutant ⇒ v malloc nothing
⇒ access v ⇒ fail

return NULL

17. Assume that we have the procedure in Figure 1 to test. Please write down a mutant with the Scalar Variable Replacement (SVR) and write down a test case (values for a, b, and c) to kill the mutant. You must explain correctly why the mutant is killed by your test case. (5pts/80)

mutant

```
v = malloc((b) * sizeof ...)
```

no mutant

(a, b, c) = (3, 1, 1)

⇒ v malloc 1

⇒ v access from 0 ~ 3

⇒ fail to access

return v